



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (Semester I) - May, 2022
STA1013 - Introduction to Statistics (3 Credits)

Answer **ALL** questions.

Time allowed: **TWO** hours.

Instructions

- This is a closed book examination.
- You are permitted to use a Scientific calculator during the examination.
- Use the answer booklets to write your answers and the single sheets for rough work.
- For any unacceptable behaviour, including the violation of examination regulations during the examination, official actions will be taken under Section 9 (Regulations Related to Exam Procedures) of the student handbook.

1. (a) Give a suitable definition of Statistics.
- (b) Briefly state the importance of statistics in Weather related studies.
- (c) The heights (in *cm*) of 30 students in a class are given below.

155	158	154	158	160	148	149	150	153	159
161	148	157	153	157	162	159	151	154	156
152	156	160	152	147	155	165	155	157	153

- i. Are these observations, Qualitative or Quantitative?
 - ii. What scale of measurements does this data belong?
 - iii. Prepare a frequency table with equal class width by taking 155-159 as one of the class intervals.
 - iv. Append a column to the table that gives the cumulative frequency distribution.
 - v. Using the frequency table, calculate the percentage of students whose height is less than 165 *cm*.
- (d) If you were asked to visualize the above data set using a graph, would you choose a Pie Chart or a Histogram? Justify your answer. *R*
2. (a) The following table shows the frequency distribution of 300 telephone calls according to their duration in seconds (*s*).

Duration (<i>s</i>)	0-30	30-60	60-90	90-120	120-150	150-180	180-210
Number of calls	9	17	43	82	81	44	24

Use the above data set to calculate,

- i. the 7th decile (D_7) and 50th percentile (P_{50}) for the duration of calls.
 - ii. the standard deviation for the duration of a call.
 - iii. the Karl Pearson's Coefficient of Skewness and interpret it.
- (b) In a situation where the Karl Pearson's coefficient of skewness is not available for a data set with **more** extreme values, what is the most appropriate skewness measurement would you recommend to describe the skewness? Briefly explain.

R

[P.T.O.]

[Question (2) cont..]

(c) The standard deviation of a distribution is 3. What should be the value of the fourth moment so that the distribution is

- i. Mesokurtic
- ii. Platykurtic

(d) Consider the following data set of daily expenditure (in Rupees) of a batch of 20 students.

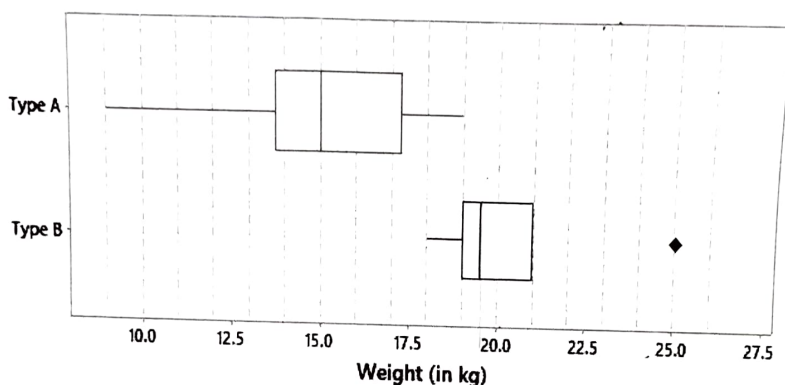
86, 92, 100, 93, 89, 95, 79, 98, 68, 62, 71, 75, 88, 86, 93, 81, 100, 86, 96, 52

The standard deviation of the expenditure is 13.2 Rupees.

- i. What percentage of data points lie within two standard deviations from the mean?
- ii. Is the result you obtained above consistent with what Chebyshev's Theorem concludes?

3. (a) List down the five-number summary statistics.

(b) Boxplots for the baggage weights (in kg) of each of the 10 passengers traveling in two types, Type A and Type B, of passenger carriers are given in the below figure.



Use the figure to answer the following questions.

- i. Write down the five-number summary statistics for the baggage weight in the Type B carrier.
 - ii. Which carrier carries the heaviest baggage?
 - iii. Calculate the inter-quartile range (IQR) for the baggage weights in the Type B carrier.
 - iv. Comment on the shape of the distribution of the baggage weights in the Type A carrier.
 - v. Do you find any outliers in the above figure? If Yes, what are they?
 - vi. Calculate the percentage of passengers in both carriers who carry baggage that weighs 15 kg or more.
 - vii. If you were asked to calculate a measure of the center for the baggage weights in each carrier, what measure of center would you recommend? Why?
- (c) Three judges have ranked the performances of eight competitors at a gymnastic competition. The data set is given in the following table.

Judge 1	1	6	5	3	2	4	7	8
Judge 2	3	5	8	4	7	2	1	6
Judge 3	6	4	8	1	2	3	5	7

Use an appropriate correlation coefficient to determine which pair of judges has the most similarity when evaluating the performances. Show all your calculations.

$$1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

[Question (3) cont..]

(d) Draw example scatter diagrams for the following correlation coefficients (r).

i. $r = -1$

ii. $r = 0.3$

4. (a) Given the regression equations of Y on X and X on Y are $Y = X$ and $4X - Y = 3$, respectively, find the value of the coefficient of correlation r . (A)
- (b) Assuming that there is a linear relationship between the height (X in kg), and weight (Y in cm), use the summary statistics given in the following table to estimate the height of an individual whose weight is 75 kg . (A)

	Height in cm	Weight in kg
Mean	165	80
Standard deviation	8	10
Correlation coefficient	0.6	

- (c) i. Briefly explain, What is an Ideal index number?
ii. Give an example for an Ideal index number.
- (d) The prices of three groups of commodities for the years 2011 to 2015 are given in the following table.

Group	2011	2012	2013	2014	2015
A	5	6	9	10	12
B	8	10	15	20	25
C	16	20	24	30	36

Calculate the chain base index numbers with 2011 as the base year.

- End of Paper -

$$B^s = \frac{Y - \bar{Y}}{\bar{Y}}$$
$$Y - \bar{Y}$$